ARANYA CHATTERJEE

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EXPERIENCE

Machine Learning Intern — Warewolf *May 2026 – Present*

- Collaborated with a cross-functional engineering team to build and refine a **YOLO-based object detection** pipeline for identifying objects in surveillance images and video footage.
- Annotated and curated a dataset of ~780 surveillance images using **Roboflow**, applying bounding-box labeling and class organization to improve training data quality.
- Performed data cleaning and preprocessing (filtering, deduplication, normalization) on surveillance datasets to boost model training reliability.
- Used **OpenCV** for frame extraction, image preprocessing, and visualization of detection outputs, applying core deep learning and CV concepts across the model training and evaluation cycle.

EDUCATION

Madhav Institute of Technology and Science, Gwalior *2023 – 2027*

Bachelor of Technology — Computer Science Engineering | CGPA: 7.7

St. Jude's School, Shahdol — CBSE (Class XII), 86% *2022*

D.A.V Burhar, Burhar — CBSE (Class X), 91% *2020*

PROJECTS

MultimodalRAG — Intelligent Video & Document Assistant *January 2026*

Tools: Python, LangChain, Whisper AI, FAISS, Groq API, MoviePy, Streamlit, Sentence Transformers | GitHub

- Engineered a multimodal RAG system processing video (MP4, MOV, AVI) and documents (PDF, TXT, DOCX, CSV) via integrated transcription and semantic search, using Whisper AI and FAISS for cross-modal retrieval with sub-second response times and 95% retrieval accuracy.
- Built a Streamlit interface with dark-theme UI and real-time AI-powered Q&A with source citations, reducing video processing time by 40% while supporting files up to 100MB.

"MASTERJI" — RAG-Powered Chatbot *November 2025*

Tools: Python, LangChain, Hugging Face, FAISS, Llama-2, Streamlit | GitHub

- Architected a Retrieval-Augmented Generation (RAG) system using Python and LangChain, achieving a 30% improvement in response accuracy over base LLM outputs.
- Integrated FAISS vector embeddings for semantic document retrieval with sub-second query response and reduced hallucinations.
- Developed a Streamlit interface supporting multi-format document uploads (PDF/TXT) with real-time Q&A, fully sourced citations, and context-aware generation via Llama-2.

AI Finance Assistant *June 2025*

Tools: Python, Scikit-learn, Streamlit, ML Models (Logistic/Linear Regression) | GitHub

- Engineered a multi-model ML pipeline (Scikit-learn, Logistic/Linear Regression, TF-IDF, Exponential Smoothing) achieving 92% accuracy in expense categorization and personalized budget forecasting; added interactive dashboards driving a 15% reduction in unnecessary spending in test simulations.

TECHNICAL SKILLS

- **Programming Languages:** Python (Advanced), Java (HackerRank 5-Star), SQL (MySQL), C/C++
- **AI/ML & Computer Vision:** PyTorch, TensorFlow, Keras, Scikit-learn, LangChain, Hugging Face, YOLO, OpenCV, Roboflow, LLM Fine-Tuning, DSPy, GEPA
- **Data Science & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Streamlit, Exploratory Data Analysis (EDA)
- **Tools & DevOps:** Git, GitHub, VS Code, Jupyter Notebook

LEADERSHIP & POSITIONS OF RESPONSIBILITY

Team Leader — Code-N-Compete National Hackathon (IIT Kanpur) *April 2025*

- Spearheaded a developer team in a high-pressure national hackathon, delivering a functional prototype within a 48-hour sprint and coordinating cross-functional workflows for a 25% faster integration phase.

Second Author — Research Publication (ICMCTR 2025) *May 2025*

- Co-authored "A Hybrid IoT + AI Approach for Risk Management in Algorithmic Trading," presented at ICMCTR-2025, contributing to technical drafting and experimentation on risk mitigation for high-frequency financial systems.

CERTIFICATIONS & ACHIEVEMENTS

- **National Finalist** (Hack-O-Calypse, IIT Jammu; Code-N-Compete, IIT Kanpur); published research at ICMCTR-2025.